DFW Housing Market Analysis

Data Dictionary

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Project Type: End-to-End Data Analytics Portfolio Project
Analysis Period: May 2016 – May 2026
Tools Used: Google BigQuery, SQL, Tableau, Microsoft Excel

Table of Contents

1. Purpose
2. Project Overview
3. Dataset Inventory
4. Dataset 1 – latest_home_values.csv
5. Dataset 2 – city_appreciation.csv
6. Dataset 3 – county_appreciation.csv
7. Dataset 4 – buyer_opportunity_index.csv
8. Dataset 5 – county_performance.csv
9. Calculated Metrics
10. Data Relationships
11. Data Quality Notes
12. Summary

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1. Purpose

The purpose of this Data Dictionary is to provide a comprehensive reference for every dataset, field, and calculated metric used throughout the DFW Housing Market Analysis project.

This document promotes consistency, transparency, and reproducibility by defining each data element used in the SQL analysis and Tableau dashboards.

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2. Project Overview

This project analyzes residential housing market trends across the Dallas–Fort Worth metropolitan area to determine which cities provide the best balance of affordability, historical appreciation, and buyer opportunity.

The project consists of five analytical datasets created from the Zillow Home Value Index (ZHVI). These datasets support four interactive Tableau dashboards and the accompanying executive and technical documentation.

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3. Dataset Inventory

Dataset	Purpose
latest_home_values.csv	Current housing market values by city
city_appreciation.csv	Ten-year appreciation by city
county_appreciation.csv	Average appreciation by county
buyer_opportunity_index.csv	Composite ranking of buyer opportunities
county_performance.csv	County-level housing metrics, component rankings, and composite performance ranking

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4. Dataset 1 – latest_home_values.csv

Purpose

Provides the most recent estimated home value for each city included in the analysis.

Field Name	Data Type	Description	Example
City	Text	Name of the city	Frisco
County	Text	County where the city is located	Collin
State	Text	State abbreviation	TX
Current Home Value	Currency (Decimal)	Latest Zillow Home Value Index estimate	$645,320

Dataset Usage

Used in:

- Current Market Snapshot Dashboard
- KPI calculations
- County averages
- Highest and lowest value city analysis
- Bubble map visualization

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5. Dataset 2 – city_appreciation.csv

Purpose

Measures the percentage increase in home values for each city between May 2016 and May 2026.

Field Name	Data Type	Description	Example
City	Text	City name	Mansfield
County	Text	County name	Tarrant
Appreciation Percent	Percentage	Ten-year home value appreciation	128.47%

Dataset Usage

Used in:

- Home Value Appreciation Dashboard
- Appreciation KPIs
- Highest appreciating cities
- Lowest appreciating cities
- Geographic appreciation map

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6. Dataset 3 – county_appreciation.csv

Purpose

Provides the average appreciation percentage for each county within the study area.

Field Name	Data Type	Description	Example
County	Text	County name	Denton
Average Appreciation Percentage		Mean appreciation across cities in the county	116.82%

Dataset Usage

Used in:

- County comparison chart
- County performance analysis
- Geographic market comparisons

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7. Dataset 4 – buyer_opportunity_index.csv

Purpose

Combines affordability and appreciation into a single score that ranks cities according to overall buyer opportunity.

Field Name	Data Type	Description	Example
City	Text	City name	Forney
County	Text	County name	Kaufman
Current Home Value	Currency (Decimal)	Latest estimated home value	$348,900
Appreciation Percent	Percentage	Ten-year appreciation	142.63%
Appreciation Rank	Integer	Ranking based on appreciation	5
Affordability Rank	Integer	Ranking based on current home value	17
Buyer Opportunity Score	Decimal	Composite score combining affordability and appreciation	22.0

Field Name	Data Type	Description	Example
Buyer Opportunity Rank	Integer	Overall opportunity ranking	3

Dataset Usage

Used in:

- Buyer Opportunity Dashboard
- Opportunity KPIs
- Top Opportunity Cities
- Lowest Opportunity Cities
- Opportunity Bubble Map

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8. Dataset 5 – county_performance.csv

Purpose

Provides a county-level comparison of housing market performance across the Dallas–Fort Worth region. The dataset combines average home values, historical appreciation, Buyer Opportunity Scores, and component rankings into a composite County Performance Score and Overall Rank.

Field	Description	Expected format
County	County name	Text
Average_Home_Value	Average home value for the county	Currency/decimal
Average_Appreciation	Average 10-year appreciation for the county	Percentage
Average_Buyer_Opportunity_Score	Average Buyer Opportunity Score for the county	Decimal
Home_Value_Rank	County's rank for average home value	Integer

Field	Description	Expected format
Appreciation_Rank	County's rank for average appreciation	Integer
Buyer_Opportunity_Rank	County's rank for Buyer Opportunity Score	Integer
County_Performance_Score	Composite score calculated from the three component ranks	Decimal
Overall_Rank	Final county ranking based on the composite score	Integer

Dataset Usage

- Executive Market Performance dashboard
- County Performance Score chart
- County Performance Map
- Average Home Value by County
- 10-Year Appreciation by County
- Average Buyer Opportunity Score by County
- Executive county-level KPI calculations

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9. Calculated Metrics

Average Home Value

Description: Calculates the mean current home value across all analyzed cities.

Business Purpose: Establishes a regional benchmark for comparison.

Highest Home Value

Description: Identifies the city with the highest current home value.

Business Purpose: Highlights premium housing markets.

Lowest Home Value

Description: Identifies the city with the lowest current home value.

Business Purpose: Highlights the most affordable markets.

Cities Above Average

Description: Counts cities whose home values exceed the regional average.

Business Purpose: Measures market distribution.

Average Appreciation

Description: Calculates the average percentage appreciation across all cities.

Business Purpose: Provides a benchmark for historical growth.

Highest Appreciation

Description: Identifies the city with the greatest ten-year appreciation.

Business Purpose: Highlights the strongest-performing housing market.

Lowest Appreciation

Description: Identifies the city with the lowest appreciation.

Business Purpose: Identifies slower-growing markets.

Buyer Opportunity Score

Description: Composite score combining affordability and appreciation rankings.

Business Purpose: Provides a balanced measure of investment potential for prospective buyers.

Buyer Opportunity Rank

Description: Orders cities according to Buyer Opportunity Score.

Business Purpose: Simplifies market comparison for decision-makers.

County Performance Score

Description: The average of the Home Value Rank, Appreciation Rank, and Buyer Opportunity Rank for each county.

Business Purpose: Provides a composite measure for comparing county-level housing performance across the three component rankings.

Interpretation: Lower scores indicate stronger overall performance under this model.

Formula:

$$\text{County Performance Score} = \frac{\text{Home Value Rank} + \text{Appreciation Rank} + \text{Buyer Opportunity Rank}}{3}$$

Ranking and Tie Handling: County Performance Score is calculated as the average of the Home Value Rank, Appreciation Rank, and Buyer Opportunity Rank. Overall Rank is assigned using the SQL RANK() function, ordered by County Performance Score in ascending order. Counties with identical performance scores receive the same rank. As a result, rank numbers may be skipped following a tie. In the rebuilt dataset, Tarrant County and Montague County both have a County Performance Score of 4.33 and share Overall Rank 1. This tie is retained in accordance with the existing methodology.

Overall Rank

Description: Ranks counties based on their County Performance Score, with lower scores receiving stronger rankings.

Business Purpose: Identifies the highest-performing counties under the composite model.

Interpretation: An Overall Rank of 1 represents the strongest overall result under the defined scoring method.

Tie Handling: Counties with identical County Performance Scores receive the same Overall Rank. Subsequent ranks are adjusted accordingly, so tied rankings may result in skipped rank numbers.

Important methodological distinction

The composite model treats the three component ranks equally. It does not establish that a county is the most affordable in absolute terms or guarantee that it will experience the strongest future appreciation.

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10. Data Relationships

The project datasets are connected through common geographic identifiers.

Common Fields

- City
- County

These shared fields enable consistent comparisons across datasets and support interactive filtering within Tableau.

Relationship Overview

- **latest_home_values.csv** supplies current market metrics.
- **city_appreciation.csv** provides historical growth data.
- **county_appreciation.csv** summarizes appreciation at the county level.
- **buyer_opportunity_index.csv** integrates affordability and appreciation into a unified ranking.
- **county_performance.csv** combines county-level metrics and component rankings to support the composite performance analysis.

Together, these five datasets support four Tableau dashboards that examine current market conditions, historical appreciation, buyer opportunity, and overall county performance.

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11. Data Quality Notes

Time Period

May 2016 – May 2026

Geographic Scope

Dallas–Fort Worth Metropolitan Area

Data Source

Zillow Home Value Index (ZHVI)

Validation Performed

The following validation steps were completed:

- Verified city and county consistency.
- Confirmed SQL calculations matched Tableau outputs.
- Reviewed KPI values for accuracy.
- Validated dashboard filters and interactions.
- Checked calculated rankings for logical ordering.

Assumptions

- Zillow Home Value Index values accurately represent typical residential home values.
- Historical appreciation is calculated using available monthly home value estimates.
- City-level analysis provides meaningful comparisons across the DFW metropolitan area.

Limitations

- Analysis is limited to residential home values.
- Results are based on historical market performance and should not be interpreted as forecasts.
- Market conditions may change after the analysis period.

County Performance

- County performance calculations were reviewed to verify the component ranks, composite County Performance Scores, and Overall Rank ordering.
- The published dashboard was checked against the underlying analytical dataset.

The County Performance Score is a comparative measure derived from component rankings. It reflects the defined scoring model and should not be interpreted as a forecast of future home values, investment returns, or market performance.

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12. Summary

This Data Dictionary documents the five analytical datasets, their fields, and the calculated metrics used throughout the DFW Housing Market Analysis project. The datasets support four Tableau dashboards covering current housing market conditions, historical appreciation, buyer opportunity, and overall county performance.

By standardizing field definitions and calculation methods, this document improves transparency, supports reproducibility, and promotes consistent interpretation across SQL analysis, Tableau visualizations, and executive reporting.